

Math340S23-Midterm2

Wednesday, April 12, 2023 6:30 PM



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Math 340: Elementary Matrix and Linear Algebra

MIDTERM TWO

Wednesday, April 12, 2023, 5:45pm-7:15pm

Please circle your instructor and your Discussion Session (start time given) in the table below.

Dr. Ruhui Jin	Dr. Yousheng Shi	Dr. Kaitlyn Phillipson	Dr. Jiuhua Hu
Marios, T 7:45 AM	Qi, M 8:50 AM	Joey, M 12:05 PM	Jingyi, T 11:00 AM
Ang, T 8:50 AM	Qi, M 9:55 AM	Sun Woo, M 1:20 PM	Jingyi, R 11:00 AM
Marios, R 7:45 AM	Zaidan, W 7:45 AM	Joey, W 12:05 PM	Gautam, T 12:05 PM
Ang, R 8:50 AM	Zaidan, W 8:50 AM	Sun Woo, W 1:20 PM	Gautam, R 12:05 PM
Ang, R 9:55 AM	Qi, W 9:55 AM		Jingyi, T 9:55 AM
Emma, T 11:00 AM			
Emma, T 12:05 PM			
Marios, R 11:00 AM			
Emma, R 12:05 PM			

Name (Printed): Key Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION. **DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.**

- This is a 90-minute exam. It consists of nine problems for a total of 50 points; the exam is 7 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices, P_d is the vector space of polynomials of degree at most d , $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.

1. (5 points, 1 point each) Are each of the following true or false statements? Circle your answer.
Justification is not necessary.

a.) ☒ T ☐ F Any linearly independent set of 6 matrices in M_{32} will form a basis for M_{32} . $\rightarrow \dim(M_{32}) = 6$

b.) T ☒ F P_3 and $\mathbb{R}^3$ are isomorphic vector spaces. $\dim(P_3) = 4$ $\dim(\mathbb{R}^3) = 3$

c.) ☒ T ☐ F If A is a 4×4 matrix with $\text{rank}(A) = 4$, then A is invertible. $\rightarrow 4$ pivots

d.) T ☒ F If S is a subset of a vector space V and $\mathbf{0}$ is in S , then S is linearly independent.

e.) T ☒ F Any set of 4 vectors in $\mathbb{R}^3$ will span $\mathbb{R}^3$.

$\mathbf{0}$ in the set automatically makes the set linearly dependent.

Ex: this set
 $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$
 does not span $\mathbb{R}^3$.

2. (7 points) (This problem has overflow room on next page if needed)

For this problem, use A and the reduced row echelon form of A , $RREF(A)$, to answer the following questions.

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 3 \\ 0 & 1 & 2 & -2 & -6 \\ 1 & 1 & 0 & 1 & 3 \\ 2 & 5 & 6 & 2 & 6 \end{bmatrix} \quad RREF(A) = \begin{bmatrix} \overset{x_1}{1} & 0 & \overset{\text{free}}{x_3} & \overset{\text{free}}{x_4} & \overset{\text{free}}{x_5} \\ 0 & \overset{\text{pivot}}{1} & 2 & 0 & 0 \\ 0 & 0 & 0 & \overset{\text{pivot}}{1} & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

a.) (2 points) What is the rank of A ? Explain your answer in one sentence.

3 $\rightarrow$ There are 3 pivots in $RREF(A)$.
or

There are 3 nonzero rows in $RREF(A)$

b.) (2 points) What is the nullity of A ? Explain your answer in one sentence.

2 $\rightarrow$ there are 2 free variables in $RREF(A)$

or $\# \text{columns} - \text{rank} = \text{nullity}$
 $5 - 3 = 2$.

c.) (1 point) Find a basis for the span of the column vectors of A (in other words, a basis for the column space of A).

Pick up the pivots from A :

$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \\ 2 \end{bmatrix}$$

this whole thing is not the basis!

d.) (2 points) Find a basis for the nullspace of A .

$$x_3 = r, \quad x_5 = s, \quad x_1 = 2r$$

$$x_1 = 2r, \quad x_2 = -2r$$

$$x_4 = -3s$$

$$\begin{bmatrix} 2r \\ -2r \\ r \\ -3s \\ s \end{bmatrix} = r \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

3

Basis

Overflow room for Problem 2, if needed:

3. (3 points) Let $L: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3x_2 \\ x_2 - x_1 \\ 2x_1 + x_2 \end{bmatrix}.$$

(Note: you can assume without verification that L is a linear transformation).

Find the standard matrix representing L .

$$L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \quad L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 3 \\ -1 & 1 \\ 2 & 1 \end{bmatrix}$$

→ can't choose A !

4. (5 points) Let A be a 2×2 matrix, and let W be the set of all vectors $\mathbf{v}$ in $\mathbb{R}^2$ such that $A\mathbf{v} = 3\mathbf{v}$. Is W a subspace of $\mathbb{R}^2$? Completely justify your answer.

vectors are in $\mathbb{R}^2$ →
Nonempty: $\vec{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $A\vec{0} = \vec{0} = 3\vec{0}$ ✓. $\vec{0}$ is in W .
closed under add: let $\vec{u}, \vec{v}$ be in W . $A\vec{u} = 3\vec{u}$, $A\vec{v} = 3\vec{v}$

$$A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} = 3\vec{u} + 3\vec{v} = 3(\vec{u} + \vec{v}) \quad \vec{u} + \vec{v} \text{ is in } W.$$

closed under scalar: let $\vec{u}$ be in W , c be in $\mathbb{R}$.

$$A(c\vec{u}) = cA\vec{u} = c3\vec{u} = 3(c\vec{u}) \quad \checkmark. \quad c\vec{u} \text{ is in } W.$$

Yes!! It's a subspace.

* see page 14 for an alternate proof *

5. (6 points total)

(This problem has overflow room on next page if needed)

Let B be the subset of vectors in $\mathbb{R}^3$ where $B = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y + z = 2 \right\}$ under the operations

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \oplus \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 - 2 \end{bmatrix}$$

$$c \odot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} cx \\ cy \\ cz - 2c + 2 \end{bmatrix}$$

- a. (3 points) Does there exist a zero vector $\mathbf{0}$ in B such that $\mathbf{0} \oplus \mathbf{u} = \mathbf{u} = \mathbf{u} \oplus \mathbf{0}$ for all $\mathbf{u}$ in B ? Justify your answer completely.

Try: $\vec{0} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$. $0+0+2=2 \checkmark$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \oplus \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} x+0 \\ y+0 \\ z+2-2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \checkmark$$

$$\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \oplus \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0+x \\ 0+y \\ 2+z-2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \checkmark$$

yes!

- b. (3 points) Show that $(B, \oplus, \odot)$ is closed under $\odot$ (i.e., $c \odot \mathbf{u}$ is in B for all scalars c and $\mathbf{u}$ in B).

$$c \odot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} cx \\ cy \\ cz - 2c + 2 \end{bmatrix}$$

in B , so

$$x + y + z = 2.$$

add:
(coord.)

$$\begin{aligned} & cx + cy + cz - 2c + 2 \\ &= c(x + y + z) - 2c + 2 \\ &= c \cdot \underbrace{(x + y + z)}_{=2} - 2c + 2 \end{aligned}$$

$$= 2c - 2c + 2 = 2 \quad \text{this is in } B!$$

Overflow room for Problem 5:

6. (5 points) Determine if the following set S of vectors in P_2 is linearly independent. Fully justify your answer.

$$S = \{t^2 + 6t - 1, 4t - t^2 - 3, 2t^2 - 3t + 4\}$$

create isomorphism

$$at^2 + bt + c \rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Put in matrix:

$$\begin{aligned} \textcircled{1} \quad \det \begin{bmatrix} 1 & -1 & 2 \\ 6 & 4 & -3 \\ -1 & -3 & 4 \end{bmatrix} &= 1(16-9) + 1(24-3) + 2(-18+4) \\ &= 7 + 21 + 2(-14) \\ &= 28 - 28 = 0 \leftarrow \text{since } \neq 0, \text{ this means} \\ &\quad \text{the vectors are linearly dependent.} \end{aligned}$$

or
 $\textcircled{2}$ row reduce:

$$\begin{bmatrix} 1 & -1 & 2 \\ 6 & 4 & -3 \\ -1 & -3 & 4 \end{bmatrix} \xrightarrow{\substack{-6r_1 + r_2 \rightarrow r_2 \\ r_1 + r_3 \rightarrow r_3}} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 10 & -15 \\ 0 & -4 & 6 \end{bmatrix} \xrightarrow{\substack{1/5 r_2 \rightarrow r_2 \\ -1/2 r_3 \rightarrow r_3}} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 0 & 2 & -3 \end{bmatrix}$$

$$\downarrow -r_2 + r_3 \rightarrow r_3$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

row of zeros $\Rightarrow$ nontrivial solution to lin. comb.
 so they are lin. dep!

7. (5 points) Let $S = (\mathbf{v}_1, \mathbf{v}_2)$, and $T = (\mathbf{w}_1, \mathbf{w}_2)$ be ordered bases for P_1 , where

$$\mathbf{w}_1 = t - 1, \mathbf{w}_2 = 2t + 2.$$

If the transition matrix from S to T is $\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$, determine S (i.e. find $\mathbf{v}_1$ and $\mathbf{v}_2$). Show all work.

$$[\vec{v}_1]_T = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \rightsquigarrow \vec{v}_1 = 3(t-1) + 2(2t+2) \\ = 3t - 3 + 4t + 4 = \boxed{7t + 1}$$

$$[\vec{v}_2]_T = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightsquigarrow \vec{v}_2 = 2(t-1) + 1(2t+2) \\ = 2t - 2 + 2t + 2 = \boxed{4t}$$

8. (6 points total) Each of the following statements is **false**. Show each statement is false by providing explicit counterexamples.

a.) (2 points) The set of 2×2 matrices with zero determinant is a subspace of M_{22} .

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$1 \cdot 0 = 0 \checkmark$ $0 \cdot 1 = 0 \checkmark$ $1 \cdot 1 = 1 \times$
 $\downarrow$
not zero.

b.) (2 points) The function $L: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $L\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right) = \begin{bmatrix} u_1 \\ u_2 + 4 \end{bmatrix}$ is a linear transformation.

$$L\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = L\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 7 \end{bmatrix} \quad \leftarrow \text{not equal.}$$

$$L\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) + L\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 11 \end{bmatrix}$$

c.) (2 points) Let V be the set of real numbers ^{not $\mathbb{R}^2$} with the operation $\oplus$, where $a \oplus b = a - b$.
Then $\oplus$ satisfies the vector space property $\mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w}) = (\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w}$ for any $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .

$$1 \oplus (2 \oplus 3) = 1 - (2 - 3) = 1 - (-1) = 2$$

$$\stackrel{\text{vs}}{(1 \oplus 2) \oplus 3} = (1 - 2) - 3 = (-1) - 3 = -1 - 3 = -4$$

not equal.

9. (8 points total) (This problem has parts (a), (b), and (c))

Let $L : P_2 \rightarrow \mathbb{R}^2$ be the function defined by

$$L(at^2 + bt + c) = \begin{bmatrix} c - a \\ 2a + b \end{bmatrix}.$$

a.) (2 points) Show that L is a linear transformation.

$L(\vec{u} + \vec{v})$

$$\begin{aligned} L(a_1 t^2 + b_1 t + c_1 + a_2 t^2 + b_2 t + c_2) &= L((a_1 + a_2)t^2 + (b_1 + b_2)t + (c_1 + c_2)) \\ &= \begin{bmatrix} c_1 + c_2 - (a_1 + a_2) \\ 2(a_1 + a_2) + (b_1 + b_2) \end{bmatrix} = \begin{bmatrix} c_1 - a_1 \\ 2a_1 + b_1 \end{bmatrix} + \begin{bmatrix} c_2 - a_2 \\ 2a_2 + b_2 \end{bmatrix} \\ &= L(a_1 t^2 + b_1 t + c_1) + L(a_2 t^2 + b_2 t + c_2). \quad \checkmark \\ &= L(\vec{u}) + L(\vec{v}) \end{aligned}$$

let r be in $\mathbb{R}$.

$L(r(a_1 t^2 + b_1 t + c_1)) = L(r a_1 t^2 + r b_1 t + r c_1)$

$L(r\vec{u})$

$$= \begin{bmatrix} r c_1 - r a_1 \\ 2r a_1 + r b_1 \end{bmatrix} = r \begin{bmatrix} c_1 - a_1 \\ 2a_1 + b_1 \end{bmatrix} = r L(a_1 t^2 + b_1 t + c_1). \quad \checkmark$$

$r L(\vec{u})$

b.) (3 points) Find a basis for the kernel of L . Show all work.

Need: $\begin{bmatrix} c - a \\ 2a + b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{\substack{2r_1 + r_2 \\ -r_1 \rightarrow r_1}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$

$\begin{matrix} a & b & c \\ \text{free} \end{matrix}$

$c = r$
 $a = r$
 $b = -2r$

ker: $rt^2 - 2rt + r$

$\underbrace{r(t^2 - 2t + 1)}_{\text{Basis.}}$

(Problem 9 continued) More space for part (b), if needed:

c.) (3 points) Is $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ in the range of L (also known as the image of P_2 under L)? Fully justify your answer.

① $\ker$ has $\dim 1$.
 $\dim \text{range}(L) = 3 - 1 = 2 = \dim(\mathbb{R}^2)$.
 so L is onto, so yes, $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ is in the range of L .

② let $L(at^2 + bt + c) = \begin{bmatrix} c-a \\ 2a+b \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & 3 \\ 2 & 1 & 0 & -2 \end{array} \right] \xrightarrow{2r_1 + r_2} \left[\begin{array}{ccc|c} -1 & 0 & 1 & 3 \\ 0 & 1 & 2 & -2 \end{array} \right] \xrightarrow{-r_1 + r_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & -3 \\ 0 & 1 & 2 & -2 \end{array} \right]$$

this system is consistent.
yes.

Scratch paper page!!

Problem 4 alt: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ If $\begin{bmatrix} x \\ y \end{bmatrix}$ is in ω , $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 3 \begin{bmatrix} x \\ y \end{bmatrix}$

$$\begin{bmatrix} ax+by \\ cx+dy \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

let $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ be in ω .

$$A(\vec{u} + \vec{v}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix} = \begin{bmatrix} a(x_1+x_2)+b(y_1+y_2) \\ c(x_1+x_2)+d(y_1+y_2) \end{bmatrix} = \begin{bmatrix} \overbrace{ax_1+by_1}^{3x_1} + \overbrace{ax_2+by_2}^{3x_2} \\ \underbrace{cx_1+dy_1}_{3y_1} + \underbrace{cx_2+dy_2}_{3y_2} \end{bmatrix}$$

$$= \begin{bmatrix} 3x_1+3x_2 \\ 3y_1+3y_2 \end{bmatrix} = 3 \begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix} \checkmark. \quad \text{closed under add.}$$

let $\begin{bmatrix} x \\ y \end{bmatrix}$ be in ω , r in $\mathbb{R}$.

$$A(r\vec{u}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} rx \\ ry \end{bmatrix} = \begin{bmatrix} arx+byr \\ crx+dry \end{bmatrix} = r \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix} = r \begin{bmatrix} 3x \\ 3y \end{bmatrix} = 3 \begin{bmatrix} rx \\ ry \end{bmatrix} \checkmark.$$

$3(r\vec{u})$

closed under
scalar mult.

ω is a subspace!